An Index Obstruction to Renorming Weighted Bicyclic Algebras

Automated mathematics run `fa_banach_001`, lane 7

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Status. Full solution, likely valid, pending human review. The source's renorming question has a negative answer: for its weighted bicyclic algebra $\mathcal{A}_x$, every equivalent unital Banach-algebra norm has $C_{\mathrm{DI}} \geq e^{2x}$. The original weighted norm attains equality.

1 Source question and result

Let $BC = \langle p, q : pq = e \rangle$ be the bicyclic monoid. Every element has a unique reduced form $q^\alpha p^\beta$, $\alpha, \beta \geq 0$. For $x > 0$, Daws and Horváth define the weight

$$\omega_x(s) = \begin{cases} 1, & s = e \text{ or } s = qp, \\ e^{x(\alpha+\beta)}, & s = q^\alpha p^\beta \notin \{e, qp\}, \end{cases}$$

and the unital Banach algebra $\mathcal{A}_x = \ell^1(BC, \omega_x)$. For a unital Banach algebra A they set

$$C_{\mathrm{DI}}(A) = \inf\{\|a\|\|b\| : ab = 1,\ ba \neq 1\}.$$

They prove that the standard witnesses δ_p, δ_q stay large under every equivalent norm, but leave open whether some other witnesses can make $C_{\mathrm{DI}}(\mathcal{A}_x)$ equal to 1, or bounded by an absolute constant [1, p. 18]:

> We have been unable to decide if it is possible to renorm $\mathcal{A}_x$ so as to get $C_{\mathrm{DI}}(\mathcal{A}_x, \|\cdot\|_0) = 1$ (or just be smaller than some absolute constant). The difficulty is that, given the previous proposition, we need consider the possibility of some other elements $c, d \in \mathcal{A}_x$ with $cd = 1$ and $dc \neq 1$, while also considering an arbitrary renorming.

Figure 1: The renorming question in [1, p. 18].

Theorem 1 (Renorming obstruction)**.** *Let $x > 0$. If $\|\cdot\|_0$ is any norm equivalent to the original norm of $\mathcal{A}_x$ for which $(\mathcal{A}_x, \|\cdot\|_0)$ is a unital Banach algebra with $\|1\|_0 = 1$, then*

$$\boxed{C_{\mathrm{DI}}(\mathcal{A}_x, \|\cdot\|_0) \geq e^{2x}.} \tag{1}$$

For the original weighted norm, equality holds. Hence there is no absolute constant K such that every Dedekind-infinite Banach algebra admits an equivalent unital Banach-algebra norm with $C_{\mathrm{DI}} \leq K$.

The proof combines annular characters with the Fredholm index of a Toeplitz operator. The index forces every possible one-sided inverse witness—not only the displayed generators—to wind nontrivially around the annulus.

2 Annular characters

For z in the closed annulus

$$\mathbb{A}_x = \{z \in \mathbb{C} : e^{-x} \le |z| \le e^x\},$$

define on the monoid basis

$$\chi_z(\delta_{q^\alpha p^\beta}) = z^{\alpha-\beta}. \tag{2}$$

The degree $\alpha - \beta$ is additive under multiplication in BC, so χ_z is multiplicative. Moreover

$$|z^{\alpha-\beta}| \le e^{x|\alpha-\beta|} \le \omega_x(q^\alpha p^\beta);$$

the exceptional elements e, qp both have degree zero. Thus (2) extends to a continuous character of $\mathcal{A}_x$.

For $a \in \mathcal{A}_x$ write $\widehat{a}(z) = \chi_z(a)$. The defining series converges uniformly on $\mathbb{A}_x$ and normally in its interior, so $\widehat{a}$ is continuous on the closed annulus and holomorphic inside. Since equivalent Banach-algebra norms do not change invertibility or spectra,

$$\|a\|_0 \ge r_{\mathcal{A}_x}(a) \ge \sup_{z\in\mathbb{A}_x} |\widehat{a}(z)|. \tag{3}$$

3 Every one-sided inverse has nonzero index

Let S be the unilateral shift on $\ell^2(\mathbb{N}_0)$ and define

$$\pi(\delta_q) = S, \qquad \pi(\delta_p) = S^*. \tag{4}$$

Because $S^*S = I$, this is a contractive representation of the bicyclic monoid and extends continuously to $\mathcal{A}_x$. Its image lies in the Toeplitz algebra $\mathcal{T} = C^*(S)$, and the Toeplitz symbol of $\pi(a)$ is

$$\sigma(\pi(a))(\zeta) = \widehat{a}(\zeta), \qquad |\zeta| = 1. \tag{5}$$

We record that π is faithful. Indeed, suppose $a = \sum_{\alpha,\beta} c_{\alpha\beta}\delta_{q^\alpha p^\beta}$ and $\pi(a) = 0$. Fix an integer $d = \alpha - \beta$. Along the dth matrix diagonal, the entry beginning at column j is the absolutely convergent partial sum

$$\sum_{\substack{\beta\le j\\ \beta+d\ge 0}} c_{\beta+d,\beta}.$$

All these sums vanish; taking successive differences gives $c_{\beta+d,\beta} = 0$ for every admissible β. Varying d recovers all coefficients.

Now take arbitrary $a, b \in \mathcal{A}_x$ with

$$ab = 1, \qquad ba \ne 1. \tag{6}$$

Set $A = \pi(a)$ and $B = \pi(b)$. Then $AB = I$, while faithfulness gives $BA \ne I$. Hence A is surjective but not injective. On the other hand,

$$\widehat{a}(z)\widehat{b}(z) = 1 \quad (z \in \mathbb{A}_x),$$

so the Toeplitz symbol of A is nowhere zero on the unit circle. Therefore A is Fredholm, and

$$\operatorname{ind} A = \dim\ker A > 0.$$

The Toeplitz index theorem [2, Chapter 7] gives

$$\operatorname{ind} A = -\operatorname{wind}(\widehat{a}|_{\mathbb{T}}, 0). \tag{7}$$

In particular the integer $k = \operatorname{wind}(\widehat{a}|_{\mathbb{T}}, 0)$ is nonzero.

4 Winding forces a large annular condition number

Lemma 1 (Annular oscillation)**.** *Let f be continuous and nonvanishing on $\mathbb{A}_x$, holomorphic in its interior, and let its winding number around zero be k. Then*

$$\left(\sup_{\mathbb{A}_x} |f|\right)\left(\sup_{\mathbb{A}_x} |f|^{-1}\right) \geq e^{2x|k|}. \tag{8}$$

Proof. The function $z^{-k}f(z)$ has winding zero and hence admits a holomorphic logarithm g on the open annulus. Thus

$$\log|f(re^{it})| = k\log r + \Re g(re^{it}).$$

The circular mean of $\Re g(re^{it})$ is independent of r: this follows from the Laurent expansion of g, whose nonconstant terms have mean zero. Consequently the circular means of $\log|f|$ on the circles of radii e^x and e^{-x} differ by $2kx$. If $M = \max_{\mathbb{A}_x}\log|f|$ and $m = \min_{\mathbb{A}_x}\log|f|$, then $M - m \geq 2x|k|$. Exponentiation gives (8). □

Apply Lemma 1 to $f = \widehat{a}$. Since $\widehat{b} = 1/\widehat{a}$, equations (3) and (7) yield

$$\|a\|_0\|b\|_0 \geq \left(\sup_{\mathbb{A}_x}|\widehat{a}|\right)\left(\sup_{\mathbb{A}_x}|\widehat{b}|\right) \geq e^{2x|k|} \geq e^{2x}.$$

This holds for every pair in (6); taking the infimum proves (1). In the original weighted norm, $\delta_p\delta_q = 1$ and $\delta_q\delta_p \neq 1$, while

$$\|\delta_p\|_{\omega_x} = \|\delta_q\|_{\omega_x} = e^x.$$

Thus $C_{\mathrm{DI}}(\mathcal{A}_x, \|\cdot\|_{\omega_x}) = e^{2x}$, completing the proof of Theorem 1. □

5 Verification, scope, and novelty boundary

The proof uses only the algebra and weight defined in the source plus the standard Toeplitz index theorem. The two points that close the authors' stated gap are faithfulness of the shift representation and the fact that a nonzero index enforces annular, rather than merely unit-circle, oscillation. The conclusion is invariant under arbitrary equivalent unital Banach-algebra norms because it is expressed through characters and spectral radii.

The result answers both formulations in the quoted passage: C_{DI} cannot be renormed to 1, and no absolute bound independent of x exists. It also determines the exact value in the original norm, improving the source's lower estimate $(e^x/86)^{1/3}$ for this family.

The run indexes, exact arXiv id and title, and arXiv searches for the Dedekind-infinite renorming terminology were checked. No later answer or duplicate packet was found. Novelty confidence is moderate: the individual Toeplitz and annulus ingredients are classical, but their use to control every C_{DI} witness in this weighted algebra was not located. Human review should focus on the faithful extension of (4) and the application of the Toeplitz index theorem to arbitrary elements of $\mathcal{A}_x$.

References

[1] M. Daws and B. Horváth, *Ring-theoretic (in)finiteness in reduced products of Banach algebras*, Canad. J. Math. 73 (2021), 1423–1458; arXiv:1912.07108.

[2] R. G. Douglas, *Banach Algebra Techniques in Operator Theory*, Academic Press, 1972.